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FREE EXECUTIVE SUMMARY

POWERING AI

The U.S. Data Center Buildout Report · 2026

Inside: the 8 findings, the one chart that explains the AI power crunch, and a preview of the regional race. Updated October 2026.

Full 36-page report → gridlas.com

EXECUTIVE SUMMARY

Electricity became AI's hardest constraint

U.S. data centers used 4.7% of the nation's electricity in 2024, and Lawrence Berkeley National Laboratory projects about 12% by 2030 (range 9.5–15.3%). Compute is built in months. New power plants take a median of about five years to connect to the grid, and data centers wait in their own utility queues. This summary maps where that collision is happening.

> **474 GW queued in Texas, about 90% of it data centers.**

THE 8 FINDINGS

1 Demand is outrunning supply.
4.7% of U.S. power in 2024, about 12% by 2030, needing roughly 148 GW of new grid connections (LBNL).

2 A handful of metros dominate.
Northern Virginia runs about 4,500 MW; Dallas–Fort Worth and Atlanta have each passed 1 GW (CBRE H1 2026).

3 Texas is the breakout market.
ERCOT's large-load queue went from 63 GW (end of 2024) to about 474 GW by June 2026.

4 The queue is the real constraint.
New power plants take a median of 61 months to connect. Of 2000–2020 requests, only 13% are operating; about 75% withdrew.

5 Data centers wait in their own line.
Utility large-load queues vary by utility; in Virginia, Dominion is assigning connection dates out to 2031.

6 Gas is back.
Gas interconnection requests jumped 86% to 253 GW in 2025, while solar, storage and wind requests fell.

7 Behind-the-meter is surging.
xAI's Colossus ran 35 gas turbines (~422 MW) on an 8 MW grid connection. Now turbines are the bottleneck.

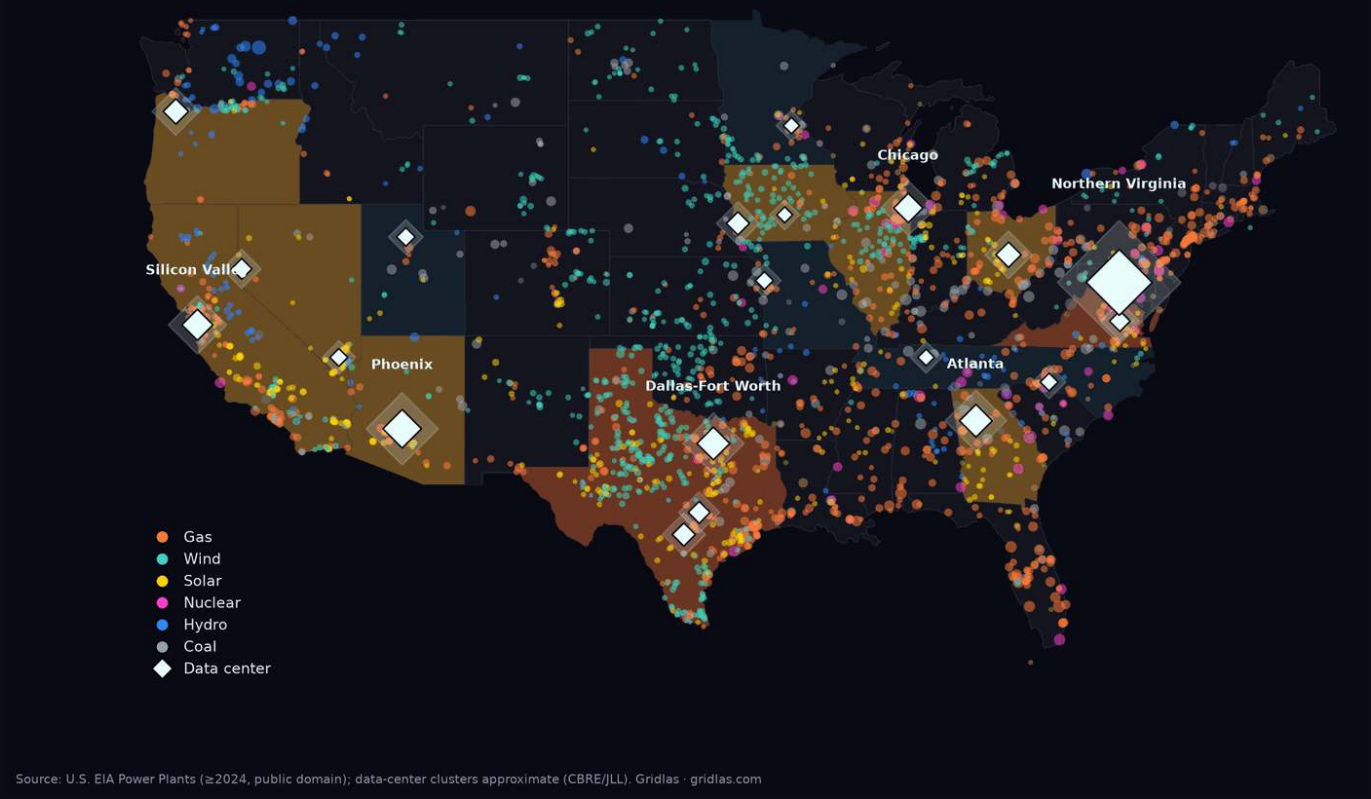
8 The next wave is moving inland.
Atlanta has about 2,880 MW under construction, the most of any U.S. market; central Ohio is adding gigawatt campuses.

THE CHART THAT EXPLAINS IT

AI is colliding with the grid

Generation vs. Demand

Supply: EIA power plants ≥100 MW (colored by fuel) · Demand: data-center clusters (white) on load-shaded states



Source: U.S. EIA Power Plants (≥2024, public domain); data-center clusters approximate (CBRE/JLL). Gridlas · gridlas.com

Supply is everywhere (2,459 utility-scale plants), but demand concentrates onto specific, already-stressed grids. The full report ranks the top 15 metros, goes deep on Texas, Virginia, Arizona, Ohio and Georgia, and closes with a 12-month outlook.

Get the full report

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Sources: LBNL, Data Center Energy Usage Report: 2025 Update (June 2026) and "Queued Up" (2026 ed.); ERCOT large-load queue (July 2026); CBRE North America Data Center Trends H1 2026; Dominion Energy GS-5 Large-Load Rate Class Report (May 2026); U.S. EIA power-plant data. Data-center cluster figures are approximate. Independent research, unaffiliated. © 2026 Gridlas.